

## **Healthcare Encounter Performance Analytics**

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# Healthcare Encounter Performance Analytics

*Power BI ETL, star-schema modeling, DAX analysis, and interactive reporting*

|                            |                                        |                         |                         |                                      |
|----------------------------|----------------------------------------|-------------------------|-------------------------|--------------------------------------|
| <b>1,500</b><br>Encounters | <b>479</b><br>Patients with encounters | <b>100</b><br>Providers | <b>10</b><br>Facilities | <b>2019–2021</b><br>Admission period |
|----------------------------|----------------------------------------|-------------------------|-------------------------|--------------------------------------|

## 1. Dataset Description

The supplied healthcare dataset contains eight CSV source tables: seven dimensions (Date, Diagnosis, Facility, Patient, Payer, Procedure, and Provider) and one encounter fact table. The analytical grain is one row per healthcare encounter. The fact table contains 1,500 encounters for 479 encountered patients, 100 providers, 10 facilities, five payer groups, 20 diagnosis identifiers, and 15 procedure identifiers. Admissions span 2019–2021; corrected discharge dates extend into January 2022. The report evaluates encounter volume, length of stay, readmission, charges, costs, margins, and operational variation.

## 2. Power Query / ETL Summary

| ETL stage                | Transformation and rationale                                                                                                                                                                                |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Import and rename        | Loaded all eight CSV tables and renamed them for readable report use, including Calendar Table, Patient Table, Facility Table, and Encounter Table.                                                         |
| Profile data quality     | Reviewed nulls, blank strings, duplicate rows and keys, type consistency, unusual characters, and numerical distributions. No missing values, empty strings, duplicate keys, or outliers required deletion. |
| Assign data types        | Applied Date, Whole Number, Decimal/Currency, and Text types according to each field's semantic role before loading the model.                                                                              |
| Enrich the calendar      | Added Start of Year, Start of Quarter, Start of Month, Month Name, and Week of Year; used Start of Month for correctly ordered multi-year visuals.                                                          |
| Create analytical fields | Combined Gender, AgeGroup, and State into Patient Profile; converted ReadmissionFlag values of 1 and 0 into Yes and No categories.                                                                          |
| Repair discharge dates   | Retained the original discharge key, recalculated discharge date as Admission Date + Length of Stay, and created a corrected relationship key.                                                              |
| Document and load        | Used descriptive Applied Step names, verified row counts after transformation, and loaded only analysis-ready tables into Power BI.                                                                         |

**Critical quality resolution 398** of 1,500 source DischargeDateID values were invalid because calendar keys had been changed by integer addition. Recalculating from the admission date and length of stay produced valid dates, preserved auditability, and identified 12 discharges in 2022.

### 3. Star-Schema Data Model

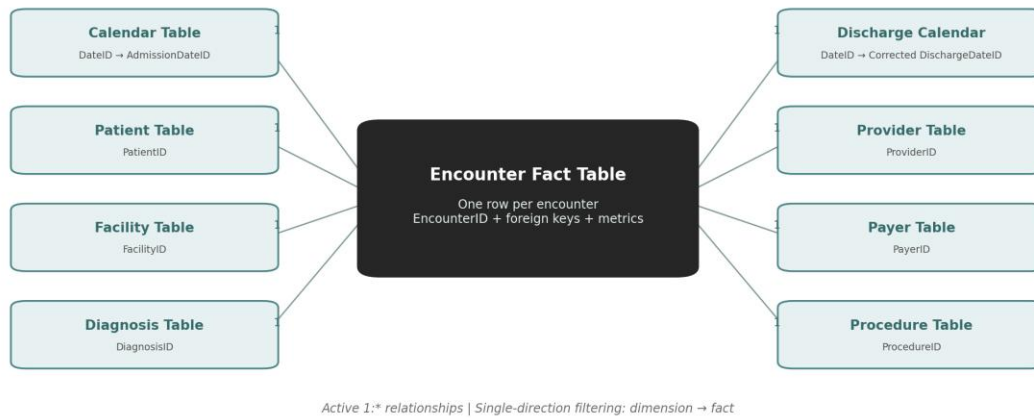
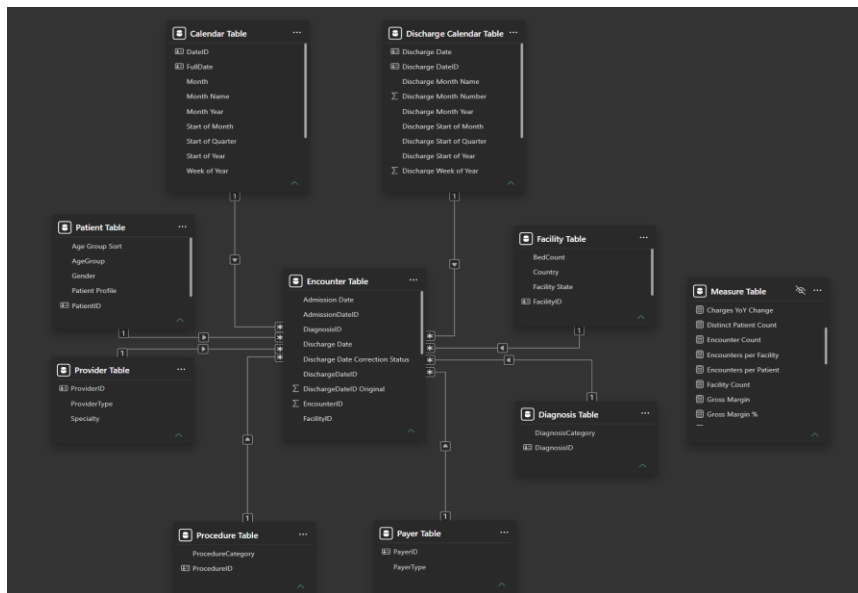


Figure 1. Optimized star schema. All relationships are active 1:\* and filter from dimension to fact.



Each dimension key is unique on the one side and connects to its corresponding foreign key in the Encounter Fact Table. Calendar Table filters AdmissionDateID, while a separate Discharge Calendar Table filters the corrected DischargeDateID. Relationships are active, one-to-many, and single direction; no bidirectional, many-to-many, or dimension-to-dimension paths were introduced. A validation matrix combining fields from every source returned the expected annual encounter counts of 476, 534, and 490, totaling 1,500.

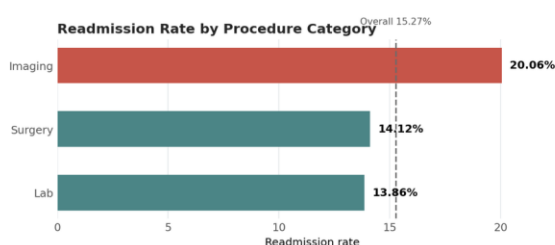
## 4. DAX Measure Catalogue

Measures were centralized in a dedicated Measures Table and grouped into volume, utilization, financial, readmission, and time-intelligence folders.

| Measure                      | Purpose / logic                                               | Measure                     | Purpose / logic                                                  |
|------------------------------|---------------------------------------------------------------|-----------------------------|------------------------------------------------------------------|
| Encounter Count              | Counts rows in the encounter fact table.                      | Non-Readmission Count       | Uses CALCULATE to count encounters where ReadmissionFlag = 0.    |
| Distinct Patient Count       | Distinct-counts PatientID in the current filter context.      | Readmission Rate            | Divides readmission count by encounter count.                    |
| Encounters per Patient       | Divides encounters by distinct encountered patients.          | Overall Readmission Rate    | Removes filters to retain the 15.27% global benchmark.           |
| Total Inpatient Days         | Sums LengthOfStay across filtered encounters.                 | Readmission Rate Comparison | Uses SWITCH to classify above, below, or equal to overall.       |
| Average Length of Stay       | Calculates mean LengthOfStay.                                 | Readmission Variance (pp)   | Returns the percentage-point gap from overall.                   |
| Median Length of Stay        | Calculates the median, reducing sensitivity to extremes.      | Total Charges YTD           | Uses TOTALYTD over the admission calendar.                       |
| Total Charges                | Sums TotalCharges.                                            | Total Charges Previous Year | Uses SAMEPERIODLASTYEAR with a blank guard.                      |
| Total Cost                   | Sums TotalCost.                                               | Charges YoY Change          | Subtracts prior-year charges from current charges.               |
| Gross Margin                 | Subtracts total cost from total charges.                      | Charges YoY %               | Divides the year-over-year change by prior-year charges.         |
| Gross Margin %               | Divides gross margin by total charges.                        | Facility Count              | Distinct-counts facilities represented in filtered encounters.   |
| Average Charge per Encounter | Divides total charges by encounter count.                     | Provider Count              | Distinct-counts providers represented in filtered encounters.    |
| Average Cost per Encounter   | Divides total cost by encounter count.                        | Encounters per Facility     | Divides encounters by distinct facility count.                   |
| Readmission Count            | Uses CALCULATE to count encounters where ReadmissionFlag = 1. | Charges Waterfall Value     | Uses first-year charges and later YoY changes for the waterfall. |

## 5. Actionable Findings

The following findings describe associations in the supplied data. They identify where operational review should begin; they do not establish clinical causation.



| Payer Performance Snapshot |                 |                        |                |                  |
|----------------------------|-----------------|------------------------|----------------|------------------|
| PayerType                  | Encounter Count | Total Charges          | Gross Margin % | Readmission Rate |
| Employer                   | 280             | \$6,525,568.00         | 38.76%         | 13.21%           |
| Medicaid                   | 317             | \$7,781,078.00         | 40.08%         | 15.14%           |
| Medicare                   | 321             | \$8,198,986.00         | 40.48%         | 13.40%           |
| Private                    | 287             | \$6,711,068.00         | 39.92%         | 19.51%           |
| <b>Total</b>               | <b>1500</b>     | <b>\$36,294,228.00</b> | <b>39.61%</b>  | <b>15.27%</b>    |

Figure 2. Procedure readmission rates and payer performance. Source: Power BI measures and dashboard output.

**1. Prioritize imaging encounter review.** Imaging recorded a 20.06% readmission rate, 4.80 percentage points above the 15.27% overall rate; Surgery and Lab were 14.12% and 13.86%. **Action:** Audit discharge instructions, follow-up scheduling, coding, and case mix for imaging encounters before designing an intervention.

**2. Target the Private payer segment for case review.** Private encounters had the highest payer readmission rate at 19.51%, 4.25 points above overall. The Imaging–Private combination reached 25.86%, but represents 58 encounters. **Action:** Review these cases first while treating the subgroup result as a signal, not a population-wide conclusion.

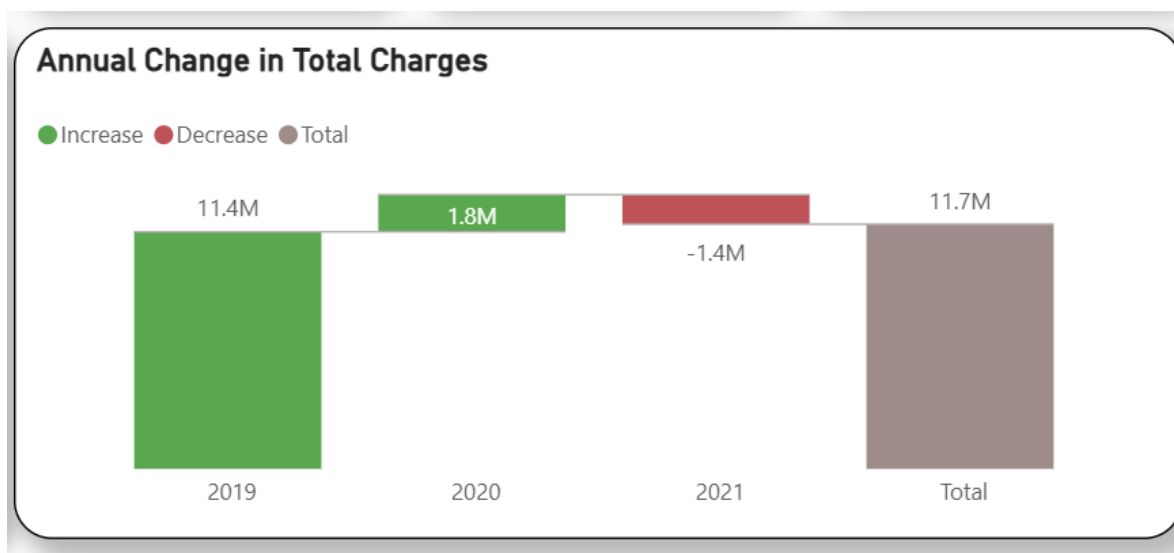


Figure 3. Annual movement in total charges. Source: Financial Performance dashboard.

**3. Investigate the 2020–2021 change.** Charges rose 15.75% to \$13.17 million in 2020, then declined 10.83% to \$11.74 million in 2021. The readmission rate also decreased from 17.42% to 12.45%. **Action:** Compare service mix, discharge practice, and follow-up processes across the two years to identify repeatable operational changes.

## 6. Facility and Provider Finding

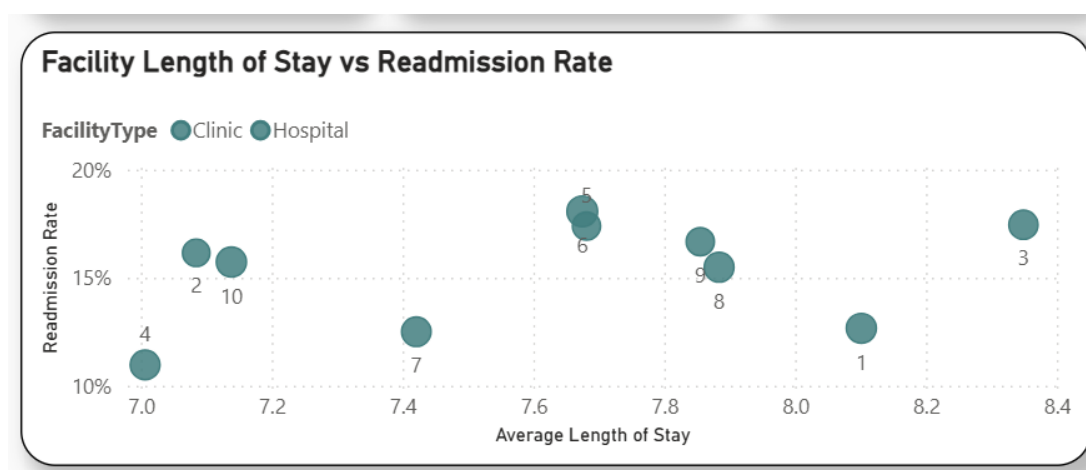


Figure 4. Facility length of stay versus readmission rate. Source: Facility & Provider Operations dashboard.

**4. Use the facility gap as a benchmarking opportunity.** Facility 6 had the highest facility readmission rate at 18.07% across 166 encounters, while Facility 4 recorded 10.97% across 155 encounters—a 7.10-point gap. **Action:** Compare transition-of-care and follow-up workflows at the two facilities, then confirm whether the difference remains after severity and case-mix adjustment.

## 7. Limitations

**Risk adjustment.** No comorbidity, severity, acuity, mortality, or socioeconomic variables were available, so facility and subgroup comparisons are descriptive.

**Synthetic-looking patterns.** Specialty-to-age combinations and clinic bed counts are not always clinically plausible; these fields should be validated before operational use.

**Financial interpretation.** Charges are not necessarily collected revenue, and gross margin is a simplified charges-minus-cost measure.

**Scope and sample size.** The dataset covers 1,500 encounters over three admission years; small intersectional groups can produce unstable percentages.

## 8. Optimization Applied

**Model optimization:** The model uses a true star schema with active 1:\* single-direction relationships and separate Admission and Discharge calendar dimensions. This removes ambiguous filter paths, avoids repeated inactive-date logic, and keeps DAX calculations simpler and more efficient.

## 9. Conclusion

The completed Power BI solution combines documented ETL, an optimized model, reusable DAX, and four interactive pages. Priorities are imaging and Private-payer readmissions, the 2020–2021 shift, and benchmarking Facility 6 against Facility 4. Validate cases and add risk adjustment before clinical or policy decisions.